

ABHITESH SHUKLA

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EDUCATION

Indian Institute of Technology, Patna <i>Bachelor of Technology in Electrical and Electronics Engineering</i> <i>With minor in Business Studies</i>	Current CPI-7.9/10 2023–2027 (Expected)
Woodland Academy, Gorakhpur <i>Senior Secondary–CBSE Board</i>	94.8% 2023
Siddharth Public School, Siddharth Nagar <i>Secondary–CBSE Board</i>	95.8% 2021

PROJECTS

SupplyShield|Python, SQL, OR-Tools, FastAPI, Streamlit | Code

- Developed a supply-chain disruption planning system to model supplier outages, capacity constraints, procurement budgets, inventory availability, and customer order deadlines.
- Built a two-stage **OR-Tools CP-SAT optimization engine** to maximize priority-weighted order fulfillment while minimizing procurement cost under real-world operational constraints.
- Designed an interactive decision-support platform using **Streamlit and FastAPI**, featuring scenario analysis, stress testing, supplier-network visualization, and reproducible result exports.

High-Resolution Semantic Segmentation | Python, TensorFlow, Jupyter Notebook | Code

- Developed a deep learning pipeline for pixel-wise segmentation of massive (6000×6000) images using a custom Residual U-Net with multi-rate dilated convolutions.
- Engineered a highly optimized, loop-free inference engine for fast patch-to-tile reconstruction, solving core memory/speed bottlenecks found in high-resolution image analysis.
- Applied advanced computer vision techniques including Test-Time Augmentation, Jaccard distance loss, and One-Cycle learning rate scheduling to maximize classification accuracy.

TECHNICAL SKILLS

Languages: Python, C++

Frameworks and Technologies: NumPy, Pandas, Matplotlib, Seaborn, Scikit-learn, PyTorch, TensorFlow, JAX

Key Courses Taken: Control System, Modern Control Theory, Introduction to AI/ML, Digital Image Processing, Deep Learning VSS, IoT

ACHIEVEMENTS AND EXPERIENCE

Market Analyst Intern | *Furnixio* 2026

- Conducted market research and competitor analysis to identify trends and growth opportunities.

Financial Analyst Intern | *Adithya Automotive Applications Limited* 2025

- Implemented customer behavior prediction models and dashboards (Power BI/Tableau).

LEADERSHIP & EXTRACURRICULAR

Overall Fest Coordinator | *CELESTA* IIT Patna Techno-Management Fest

- Led end-to-end planning, coordination, and execution of Celesta as Overall Fest Coordinator.
- Oversaw teams, operations, budgeting, logistics, and stakeholder management.

General Secretary | *Technical Affairs* Students Gymkhana, IIT Patna

- Led 17 technical clubs and 10+ institute-wide technical initiatives.
- Handled accounts, expense tracking, approvals, and reimbursement processes.